Jesus Mancilla

Senior Applied Scientist | AI Evaluation & GenAI Quality Systems

Evaluation-driven development: offline benchmarks, online monitoring, HITL scoring, and LLM-as-judge pipelines

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Professional Summary

Senior applied scientist and platform builder focused on **evaluation systems for AI-driven experiences**. Builds offline benchmark suites, automated regression tests, and human-in-the-loop scoring loops (including **LLM-as-judge** and verifier-driven refinement) to support eval-driven development. Delivered reliability and performance gains, including ~**33% hallucination reduction** and **p95 TTFT 4.3s to under 2s** (internal), and reduced open-ended analysis from ~**30h to under 8h** and document workflows from ~**90m to under 5m**. Strong cross-functional partner, known for reusable tooling, clear documentation, and adoption. Bilingual (English/Spanish).

Technical Stack & Domains

AI Evaluation & Quality: Offline benchmarks, online monitoring, acceptance tests, curated benchmark datasets, label taxonomy refinement, human-in-the-loop review, LLM-as-judge, prompt and evaluation criteria iteration
Programming: Python, SQL, JavaScript, TypeScript, Git
LLM Systems: Agents (LangGraph), RAG, vector search, multi-agent orchestration, observability and monitoring
Data & Experimentation: Regression, ANOVA, t-tests, online experiments (A/B testing), longitudinal tracking, large-scale log analysis, KPI definition
Frameworks: LangChain, LangGraph, FastAPI, React, Next.js
Domains & Scale: Consumer media (500M+ MAU), TV/streaming (~70M devices), education, commerce, automotive/HF

Work Experience

Argomai — Houston, TX (Remote)
Senior Applied Scientist — Jan 2025 – Present

- Built and governed enterprise **AI evaluation and deployment processes**, defining review loops, monitoring expectations, and quality gates across client projects.
- Authored and open-sourcing a **multilingual QA evaluation pipeline** (self-consistency → critique → refinement) with judge-centric scoring and deterministic audit checks.
- Designed reusable **platform components** (embeddings, prompt templates, orchestration SDK) and agent workflows to standardize evaluation and reuse across initiatives.
- Reduced document workflow cycle time from ~**90m to under 5m** and PM reporting from **6h/wk to under 1h** via retrieval automation and structured aggregation.

Meta — Houston, TX (Remote)
Senior Quantitative UX Researcher — Jan 2024 – Jan 2025

- Implemented evaluation pipeline patterns, including **LLM-as-judge** self-reflection and verifier-driven revisions, improving reliability of intermediate outputs for production research workflows.
- Designed **human-in-the-loop** expert review to refine outputs in natural language and curate few-shot examples, reducing analysis time ~**73%** (~30h to under 8h).
- Engineered multi-agent sampling (**self-consistency** with chain-of-thought) to generate and score candidate outputs across tasks, enabling hill-climbing style iteration.
- Ran **pre and post-launch monitoring** using bi-weekly surveys and **SQL**-based log analysis to track model quality changes and guide roadmap decisions at **500M+ MAU** scale.

Roku — San Jose, CA
Senior User Experience Researcher — Jan 2021 – Nov 2023

- Built a **modular survey analysis and reporting engine**, cutting weekly report generation from ~**4h to under 5m** via automation and reusable analysis logic.
- Created an **open-ended classifier prototype** (NLP, clustering) that later informed the scaled system deployed at Meta.
- Developed an **AI-powered indexed research database** enabling self-serve discovery and faster executive reporting across orgs.

Walmart Global Tech Sunnyvale, CA
Senior User Experience Researcher Aug 2019 – Nov 2020

- Owned product analytics for Sam's Club mobile, defining KPIs and dashboards that linked behavioral and business metrics to feature decisions.
- Standardized cross-team research operations using Jira and Confluence workflows to prioritize requests, track WIP, and improve delivery predictability.

Scrapworks Inc. Palo Alto, CA
Data Scientist Sep 2017 – Aug 2019

- Delivered deep learning forecasting for commodities, reducing prediction error by **60%** and improving decision support for traders.
- Initiated an NLP-based merchandise classifier and contributed to a patent application, building end-to-end data preparation and modeling workflows.

Suggestic Mexico City, Mexico
Senior User Experience Researcher Dec 2016 – Sep 2017

- Ran data-driven testing and analysis for new features, using analytics to iterate quickly and improve engagement in a consumer coaching app.
- Transitioned core product experience from conversational to graphical UI using insights from experimentation and behavioral signals.

Stanford University Stanford, CA
User Experience Researcher May 2016 – Nov 2016

- Conducted ML research on stress detection with 150+ hours of multimodal data and contributed to models reaching ~**90%** accuracy.

ITAM Mexico City, Mexico
User Experience Researcher Aug 2014 – May 2016

- Analyzed psychophysiological signals and built custom visualizations, applying ML to surface patterns and inform user-centered product decisions.

Selected Projects (AI Evaluation & ML Platforms)

Evaluation Benchmarking Suite: Built automated regression tests on a **500+ item** dataset to track adherence, tone, and reasoning; improved reliability (about **33%** hallucination reduction) and reduced p95 TTFT from **4.3s to under 2s** (internal).

Quant UX at Scale: 6-step GenAI evaluation pipeline (embeddings, self-consistency, LLM-as-judge, verifier revisions, HITL, few-shot curation), cutting time per study from ~**30h to under 8h** in a **500M+ MAU** context.

Enterprise RAG & Evaluation Architecture: Designed eval and governance patterns for enterprise AI, including a multilingual QA pipeline supporting iterative critique and refinement for quality improvement.

Modular Survey Analysis System: Production reporting engine with automated stats and classifier prototyping, reducing weekly reporting from ~**4h to under 5m**.

Education

Instituto Tecnológico Autónomo de México (ITAM)
M.S. in Computer Science (HCI/AI Focus) 2014 – 2016

Universidad de Colima
B.A. in Psychology 2009 – 2013

Selected Publications

Ramos-Rivera, R. E., Santana Mancilla, P. C., Garcia-Mancilla, J., & Gaytán-Lugo, L. S. (2025). Language models in education: Generative AI to optimize teacher performance analysis. InnovAcademica, 1(2), 74–85.

Ramos-Rivera, R. E., Garcia-Mancilla, J., Cárdenas-Villa, G. E., & Santana-Mancilla, P. C. (2024). Towards Improving Teacher Performance Assessment through Human-Centered AI-Powered Survey Analysis. Avances en Interacción Humano-Computadora, 9(1), 261–264.

Santana-Mancilla, P. C., Guerrero-Ibáñez, A., Contreras-Castillo, J., Garcia-Mancilla, J., & Anido-Rifón, L. (2025). Sustainable Urban Mobility: Leveraging Generative AI for Symmetry-Aware Traffic Light Optimization. Symmetry, 17(12), 2083.

Garcia-Mancilla, J., Ramirez-Marquez, J. E., Lipizzi, C., Vesonder, G. T., & Gonzalez, V. M. (2018). Characterizing negative sentiments in at-risk populations via crowd computing. International Journal of Data Science and Analytics.

For full list, see: jgmancilla.com/research-papers